Timothy Do

San Jose, CA | +1 (408) 644-5538 | timothydobsa@gmail.com | linkedin.com/in/do-timothy | github.com/dotimothy

Education

University of California, Irvine — Sep. 2019 – June 2023 *(Expected)*
B.S. Electrical Engineering, Specialization in Digital Signal Processing — *Irvine, CA*
- **Cumulative GPA:** 3.912/4.0, **Major GPA:** 3.924/4.0, **Accelerated Status**
- **Relevant Coursework:** Computer Architecture, Control Systems, Digital Image/Signal Processing, Electronics
- **Research Interests:** Hyperspectral Image Processing, Embedded IoT Systems, Parallel GPU Optimizations
- **Clubs and Societies:** IEEE, IPC, Eta Kappa Nu, Tau Beta Pi

Experience

Industrial Internet of Things Intern — June 2022 – Sep. 2022
Western Digital — *San Jose, CA*
- **Skills:** MySQL, Python, Raspberry Pi, Soldering, Linux, Lean Six Sigma
- Certified as a Six Sigma Yellow Belt for learning Six Sigma foundations
- Applied Industrial Toolsets to develop a Sensor Data Analytics Platform
- Presented results to the Vice President of Global Operations and other xecutives.

Undergraduate Student Researcher — January 2022 – Present
Irvine Computer Vision Laboratory at UC Irvine — *Irvine, CA*
- Researched under the direction of Professor Glenn Healey on the topic of Spectral Image Filtering detecting emissions in wildfires
- Digitized Imaging Filters into CSVs and compared them in MATLAB against certain chemical emissions.
- Generated altitude maps using QGIS & SRTM 1 Arc-Second Global

Vice President — Nov. 2021 – Present
Institute of Printed Circuits Club at UC Irvine — *Irvine, CA*
- Reserved club meeting rooms and sent weekly emails to club members regarding club events
- Created the IPC Club Website and managed IPC's social media platforms
- Designed circuit & programming workshops for IPC club members to apply in industrial settings

I.T. General Assistant — July 2021 – June 2022
UC Irvine Division of Continuing Education — *Irvine, CA*
- Helped Set-Up Desktops, Docks, Printers, Webcams, and Computers for over 300+ Employees
- Managed IPs, Subnet Masks, and DNS servers to access local networking services
- Monitored and Updated a Real-Time Web Server (DCE-OLLI) by rebuilding the Frontend/Backend using react.js

Projects

AM Radio & Amplifier | *LTSpice, Oscilloscopes, Soldering* — April 2022 – May 2022
- Led IPC @ UCI's Two-Part AM Radio Workshop, Creating the Schematic for Optimal Gain and Efficiency
- Designed Audio Amplifier using the *LM741* Op-Amp in an Inverting Configuration with a Single USB 5V Supply
- Designed Envelope Detector using the *1N4148* Diode and RC Band-Pass Filter for Demodulation

EECS195-Colorization | *Python, MATLAB* — Feb. 2022 – March 2022
- Colorized grayscale images using GAN and Auto-Encoder Neural Networks
- Gathered own datasets by taking pictures of different scenery
- Clustered pixels into superpixel subgraphs for uniform training weights

PhotoLab | *C, Javascript, Python, MATLAB* — June 2022 - Sep. 2022
- Implemented diverse image processing engines in Javascript (Web), C (IoT), and Python/MATLAB (AI)
- Programmed a Python Flask Web Server for Cloud-Based Image Processing
- Piped with HTTP Post Requests to send images & back with HTTP GET

Skills

Electronics: Raspberry Pi, Network Analysis, Cadence, LTSpice, Soldering, Oscilloscope & Multimeter Measurements
Programming Languages: Python, Java, C, CUDA, MATLAB, bash, MySQL HTML/CSS/Javascript, VHDL
Developer Tools: git, SSH, Makefile, Linux, Azure, VS Code, Visual Studio, PyCharm, Eclipse
Libraries: RPi.GPIO, socket, Flask, pandas, numpy, os, Matplotlib, OpenCV, PIL, tensorflow, pytorch